

Numerical Analysis PhD Exam, May 1998

Do any 8 of the following 10 problems.

1. This problem has four parts.

- (1) Give a careful description of a Householder transformation.
- (2) Describe how Householder transformations can be used to bidiagonalize a 5×5 matrix.
- (3) Describe the QR Iteration Algorithm with shift μ .
- (4) Describe how the QR Iteration is used to compute the SVD .

2. This problem has two parts.

- (1) Give a careful statement and proof of Gershgorin's Circle Theorem.
- (2) Determine whether or not the matrix

$$A = \begin{pmatrix} -3 & 1 & 1 \\ -1 & -4 & 2 \\ -1 & 0 & -2 \end{pmatrix}$$

has the property that each eigenvalue has negative real part.

3. Suppose it is known that every eigenvalue of a matrix A is real. Suppose the power method with shifts is to be used to approximate the largest eigenvalue of A . Explain how Gershgorin's method can be used to aid in the choice of reasonable shifts.

4. This problem has three parts.

- (1) Give a careful statement of the LU factorization theorem.
- (2) Give a careful statement and proof of the Cholesky Factorization Theorem.
- (3) Find (if possible) a Cholesky factorization of the matrix

$$A = \begin{pmatrix} 2 & 4 & 6 \\ 4 & 7 & 8 \\ 6 & 8 & 5 \end{pmatrix}.$$

5. This problem has two parts.

- (1) Give a careful statement of the Sturm interlacing property for symmetric matrices.
- (2) Determine whether or not the matrix

$$A = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 1 & 1 \\ 0 & 1 & 2 \end{pmatrix}$$

has any negative eigenvalues.

6. This problem has four parts.

- (1) Give a careful statement of the Pythagorean Theorem property for clamped cubic splines.
- (2) Give a careful statement of the integral minimization property for clamped cubic splines.
- (3) If $s_n(x)$ denotes the clamped cubic spline interpolant for a function $f(x)$ at the knots $(x_0, y_0), (x_1, y_1), \dots, (x_n, y_n)$, then give a precise statement for the error between $s_n(x)$ and $f(x)$.
- (4) If $s_n(x)$ denotes the clamped cubic spline interpolant for a function $f(x)$ at the knots $(x_0, y_0), (x_1, y_1), \dots, (x_n, y_n)$, then give a precise statement for the error between $s_n''(x)$ and $f''(x)$.

7. Let $f(x)$ be a real valued function on $\mathbb{R}$ with a continuous 3rd derivative at each point. Assume also that $f(x)$ has a simple root at $x = p$.

- (1) Show: If Newton's method is used to approximate p , then there is a $\delta > 0$ with the property that the convergence rate to p is 2nd order on the interval $[p - \delta, p + \delta]$.
- (2) Approximate a solution to the following nonlinear system using Newton's method. (Please don't iterate—just set up the procedure for the initial guess $x_0 = (1, 1, 1)^t$.)

$$\begin{aligned} 3x_1 - \cos(x_2 x_3) - 0.5 &= 0, \\ x_1^2 - 625x_2^2 &= 0, \\ e^{-x_1 x_2} + 20x_3 + 9 &= 0. \end{aligned}$$

- (3) What happens to the iteration process if the initial guess is changed to $x_0 = (1, 0, 1)^t$?

8. Consider the method $y_{k+1} = y_{k-1} + \frac{h}{2}(3f_k + f_{k-1})$ for solving the initial value equation $y' = f(x, y)$, $y(0) = \alpha$.

- (1) If the method is applied to the equation $y' = -10y$, $y(0) = 4$, then is there a step size h such that the global discretization error converges to zero as $k \rightarrow \infty$? Explain your answer.
- (2) Consider the equation $y' = xy(y-2)$ where $y(0) = 2$. Is this an ill-posed or a well-posed problem? Explain your answer. (Note that the general solution is $y(x) = \frac{2y_0}{y_0 + (2-y_0)e^{x^2}}$, when $y(0) = y_0$.)

9. This problem has three parts.

- (1) Give a careful statement of the Contraction Mapping Theorem.
- (2) Outline a proof of the Contraction Mapping Theorem.
- (3) Indicate how Aitken's method can be used to accelerate convergence.

10. This problem has two parts.

- (1) Explain Romberg's technique for numerical approximation of an integral.
- (2) Give a brief justification for the method.